SHORT EXACT SEQUENCES OF FILTERED OBJECTS EXTEND TO TERMWISE SPLIT SEQUENCES OF RESOLUTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8257-8310, following the statement at 8231-8255.

Editorial note. The repeated extension step is 'And so on' (8297) after three concrete instances with the factorization argument displayed; the final commutativity/exactness check reads off the displayed construction, recorded.

Allowed standard prerequisites (editorial).

- Strict morphisms in filtered quasi-isomorphisms. homology lemma-filtered-acyclic, cited at proof line 8261 (cross-chapter reference). Supplies that a, c and the differentials d_I, d_J are strict.
- Extension property of filtered injectives. lemma-injective-property-filtered-injective, cited at proof lines 8278, 8284 and 8296; short source lemma with complete proof (8122-8128). Supplies the three extension steps (b, delta^0, delta^1).

Source reference. Original lemma 05S1: sources/stacks-derived/pinned-original-source.tex, source lines 4509–4509. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TR: sources/stacks-derived/pinned-original-source.tex, source lines 8113–8113. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with enough injectives. Let $0 \to A \to B \to C \to 0$ be a short exact sequence in $\mathit{Fil}^f(\mathcal{A})$. Given filtered quasi-isomorphisms $A[0] \to I^\bullet$ and $C[0] \to J^\bullet$ where $I^\bullet, J^\bullet$ are complexes of filtered injective objects with $I^n = J^n = 0$ for $n < 0$, then there exists a commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A[0] & \longrightarrow & B[0] & \longrightarrow & C[0] & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & I^\bullet & \longrightarrow & M^\bullet & \longrightarrow & J^\bullet & \longrightarrow & 0 \end{array}$$

where the lower row is a termwise split sequence of complexes.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. As $A[0] \to I^\bullet$ and $C[0] \to J^\bullet$ are filtered quasi-isomorphisms we conclude that $a : A \to I^0$, $c : C \to J^0$ and all the morphisms d_I^n, d_J^n are strict, see Homology, Lemma 05S1. We are going to step by step construct the south-east and the south arrows in the following commutative diagram

$$\begin{array}{ccccccccc} B & \longrightarrow & C & \longrightarrow & J^0 & \longrightarrow & J^1 & \longrightarrow & \ldots \\ & {\scriptstyle\beta}\searrow & & {\scriptstyle c}\searrow & \downarrow{\scriptstyle\delta^0} & & \downarrow{\scriptstyle\delta^1} & & \\ \uparrow{\scriptstyle\alpha} & & {\scriptstyle b}\searrow & & {\scriptstyle\overline{b}}\searrow & & & & \\ A & \xrightarrow{a} & I^0 & \longrightarrow & I^1 & \longrightarrow & I^2 & \longrightarrow & \ldots \end{array}$$

As $A \to B$ is a strict monomorphism, we can find a morphism $b : B \to I^0$ such that $b \circ \alpha = a$, see Lemma 05TR. As A is the kernel of the strict morphism $I^0 \to I^1$ and $\beta = \mathrm{Coker}(\alpha)$ we obtain a unique morphism $\overline{b} : C \to I^1$ fitting into the diagram. As c is a strict monomorphism and I^1 is filtered injective we can find $\delta^0 : J^0 \to I^1$, see Lemma 05TR. Because $B \to C$ is a strict epimorphism and because $B \to I^0 \to I^1 \to I^2$ is zero, we see that $C \to I^1 \to I^2$ is zero. Hence $d_I^1 \circ \delta^0$ is zero on $C \cong \mathrm{Im}(c)$. Hence $d_I^1 \circ \delta^0$ factors through a unique morphism

$$\mathrm{Coker}(c) = \mathrm{Coim}(d_J^0) = \mathrm{Im}(d_J^0) \longrightarrow I^2.$$

As I^2 is filtered injective and $\mathrm{Im}(d_J^0) \to J^1$ is a strict monomorphism we can extend the displayed morphism to a morphism $\delta^1 : J^1 \to I^2$, see Lemma 05TR. And so on. We set $M^\bullet = I^\bullet \oplus J^\bullet$ with differential

$$d_M^n = \begin{pmatrix} d_I^n & (-1)^{n+1}\delta^n \\ 0 & d_J^n \end{pmatrix}$$

Finally, the map $B[0] \to M^\bullet$ is given by $b \oplus c \circ \beta : M \to I^0 \oplus J^0$. $\square$